Lyre Build Packet

Instrument-Maker Print Packet

Build packet folder: /tmp/lyre-codex-gina-r2-build-packet

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This packet is the printable companion to the build folder. Take it shopping or into the shop. Tear sheets at page breaks.

File Map

File	Purpose
design.md	Project intent, catalog metadata, assumptions, and validation plan.
bom.csv	Starter bill of materials with part categories, quantities, drawing refs, and notes.
sourcing.csv	Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.
cut-list.csv	Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.
drawing-brief.md	Manufacturing drawing and technical product sketch brief.
assembly-manual.md	Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.
validation.csv	Target/measured values, tolerance, environment, result, and tuning/build action log.
supplier-rfq.md	Supplier email/request-for-quote starter.
visual-bom-brief.md	Art direction for an image-forward visual BOM.
wolfram-starter.wl	Wolfram starter for physics, optimization, visualization, and validation.
README.md	Project artifact.
photo-shotlist.md	Project artifact.
risks.md	Project artifact.

design.md

Project intent, catalog metadata, assumptions, and validation plan.

Lyre Design

Intent

Create a small bench-buildable lyre packet that can move from concept to a first shop prototype without hiding the important string-load unknowns. The baseline is a compact 10-string plucked instrument with a flat soundboard, two arms, a crossbar/yoke, and individual tuning pins.

Readiness

L2 scaffold. This is suitable for design review, sourcing review, and first CAD layout. It is not ready for unattended fabrication.

Governing Model

Strings use the Mersenne-Taylor relationship:

f = (1 / (2 L)) * sqrt(T / mu)
mu = density * pi * (diameter / 2)^2

Use A4 = 440 Hz equal temperament for targets. Treat all string tensions as first-order estimates until the actual string material, gauge, and measured density are recorded.

Baseline Assumptions

Parameter	Baseline	Notes
String count	10	Diatonic C4-D4-E4-F4-G4-A4-B4-C5-D5-E5
Scale length	21 in nominal	May be fanned 18-23 in for ergonomics and tension
String material	Nylon or fluorocarbon	Gut is possible but not the first prototype default
Soundboard	0.125-0.160 in spruce, cedar, or birch ply	Final thickness requires tap and deflection checks
Frame stock	0.75 in hardwood	Grain must run continuously through arms where possible
Tuning hardware	Zither pins	Pilot holes and pin torque need validation

| Target total tension | 80-120 lbf | Design frame with margin before stringing |

Tension Safety Notes

• Keep the first prototype below 70 percent of estimated breaking load for all treble monofilament strings.
• Confirm the crossbar/yoke is not loaded in short grain.
• Bring strings to pitch gradually, alternating across the frame.
• Stop the build if the soundboard crowns, the yoke twists, or tuning pins creep under static load.

Known Gaps

• No measured string-gauge schedule yet.
• No CAD frame deflection check yet.
• No bridge downbearing measurement yet.
• No supplier price or stock verification yet.
• No prototype tuning data.

bom.csv

Starter bill of materials with part categories, quantities, drawing refs, and notes.

item	qty	unit	material_or_spec	estimated_cost_usd	notes
Frame hardwood blank	1	ea	maple, cherry, walnut, or similar clear hardwood, 0.75 in nominal	45	Avoid knots and short grain through arms
Soundboard panel	1	ea	spruce, cedar, or birch aircraft ply, 0.125-0.160 in	25	Prototype thickness to be validated by def
Back or rim spacer stock	1	set	hardwood or plywood strips	15	Supports soundbox depth if boxed body is
Zither tuning pins	10	ea	0.197 in or compatible lyre/harp pin	18	Pilot-hole fit and torque must be tested on
Bridge and nut stock	2	ea	hard maple, bone, or dense hardwood	8	Round contact points to reduce string brea
String set	1	set	nylon or fluorocarbon monofilament gauges TBD	35	Gauge schedule must be calculated and t
Adhesive	1	lot	wood glue suitable for instrument joints	8	Use fresh adhesive and full clamp coverag
Finish	1	lot	shellac, oil-varnish, or waterborne clear finish	15	Do not finish string contact surfaces until f

sourcing.csv

Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.

item	required_spec	search_terms	supplier_candidates	date_checked	unit_price_usd	lead_time	substitution_risk	notes
Zither pins	10 matching pins w[illegible]	[illegible]	[illegible]	TBD	TBD	TBD	Medium	Pin diameter and wood species
Monofilament strings	nylon or fluorocarbo[illegible]	[illegible]	[illegible]	TBD	TBD	TBD	High	Density and breaking strength m
Soundboard stock	quarter-sawn softw[illegible]	[illegible]	[illegible]	TBD	TBD	TBD	Medium	Tone and deflection vary by mat
Frame hardwood	clear straight-grain [illegible]	[illegible]	[illegible]	TBD	TBD	TBD	Low	Avoid runout at arms
Bridge/nut stock	dense wear-resista[illegible]	[illegible]	[illegible]	TBD	TBD	TBD	Low	Must polish smooth to prevent st

cut-list.csv

Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.

part	qty	rough_dimensions_in	final_dimensions_in	material	grain_or_orientation	operation	notes
Left arm	1	1.0 x 2.0 x 22.0	0.75 x profile x 21.0	clear hardwood	grain follows arm curve	bandsaw/CNC profile then rasp/sand	Leave extra at yoke joint until fitted
Right arm	1	1.0 x 2.0 x 22.0	0.75 x profile x 21.0	clear hardwood	grain follows arm curve	bandsaw/CNC profile then rasp/sand	Mirror left arm
Crossbar/yoke	1	1.0 x 1.5 x 16.0	0.75 x 1.0 x fitted length	clear hardwood	long grain end to end	drill pin row after layout verification	No short-grain load path
Soundboard	1	0.160 x 13.0 x 18.0	0.125-0.160 x body profile	spruce/cedar/ply	long grain vertical	template trim and brace if needed	Final thickness TBD
Bridge	1	0.5 x 0.5 x 12.0	profile TBD	hard maple or bone	long grain along string row	shape, slot, polish	Confirm spacing before slots
Nut	1	0.375 x 0.5 x 12.0	profile TBD	hard maple or bone	long grain along string row	shape, slot, polish	Slots must match final gauges

drawing-brief.md

Manufacturing drawing and technical product sketch brief.

Lyre Drawing Brief

Required first drawings:

- Full-size front elevation with string fan, bridge, nut, yoke, and pin row.
- Side section through body/soundboard showing bridge height and downbearing.
- Crossbar/yoke detail with pin spacing, pilot diameter, and edge distance.
- Arm joint detail showing grain direction and glue surface.

All drawings should label dimensions in inches, include metric equivalents for hardware, and mark TBD dimensions that depend on final string gauges.

assembly-manual.md

Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.

Lyre Assembly Manual

Tools

Bandsaw or CNC router, drill press, brad-point bits, clamps, rasps/files, sanding blocks, tuner, calipers, straightedge, and stringing wrench.

Build Sequence

1. Make a full-size layout with string path, bridge, nut, yoke, and pin row.
2. Cut frame arms and crossbar oversize, keeping grain continuous through the loaded members.
3. Dry-fit the frame and mark the actual speaking lengths.
4. Cut and fit the soundboard or soundbox panel.
5. Glue the frame and soundboard with flat cauls and full clamp support.
6. Shape and polish the bridge and nut. Do not cut final slots until the gauge schedule is confirmed.
7. Drill pin pilot holes in scrap first, then in the yoke after torque testing.
8. Install strings gradually, moving across the set instead of loading one side fully before the other.
9. Tune in stages over 24 hours and record movement in `validation.csv`.

Safety

Wear eye protection when stringing. Stand out of the string plane during first pitch-up. If a frame member moves visibly, detune before inspecting.

validation.csv

Target/measured values, tolerance, environment, result, and tuning/build action log.

check_id	area	target	method	tolerance	measured	result	action
VAL-001	String tuning	C4 through E5 target pitches at A4=440	chromatic tuner after 24 hour settle	+/- 10 cents	TBD	TBD	Adjust gauge, scale length, or bridg
VAL-002	String safety	all strings below 70 percent estimated break	calculate from supplier density and break strength	<=70 percent	TBD	TBD	Change gauge/material before full
VAL-003	Frame deflection	no permanent yoke or arm movement	measure before stringing and after 24 hours	<=0.030 in movement	TBD	TBD	Increase section, add lamination, o
VAL-004	Tuning pin hold	pins hold pitch through three tuning cycles	mark pin angle and pitch drift	<=5 cents drift per hour	TBD	TBD	Revise pilot hole or pin spec
VAL-005	Ergonomics	comfortable lap/bench playing reach	playing mockup photos and notes	no wrist strain	TBD	TBD	Revise body width/string fan
VAL-006	Finish clearance	finish does not bind pins or string slots	dry fit after finish cure	free movement	TBD	TBD	Mask or rework contact areas

supplier-rfq.md

Supplier email/request-for-quote starter.

Supplier RFQ Draft

Hello,

I am sourcing parts for a 10-string prototype lyre. Please quote the following items or recommend substitutions that meet the same use case:

• 10 matching zither or harp tuning pins with recommended pilot-hole size.
• Nylon or fluorocarbon monofilament strings suitable for C4 through E5 at approximately 18-23 in speaking length.
• Optional matching tuning wrench.

Please include material specifications, breaking strength or rated tension, unit price, minimum order quantity, lead time, and shipping options.

Thank you.

visual-bom-brief.md

Art direction for an image-forward visual BOM.

Visual BOM Brief

Create a one-page visual BOM with the assembled lyre at top, then callouts for frame hardwood, soundboard, tuning pins, strings, bridge, nut, adhesive, and finish. Use real photos or CAD renders when available. Mark any concept images as placeholders until replaced by shop or supplier images.

wolfram-starter.wl

Wolfram starter for physics, optimization, visualization, and validation.

```
(* Lyre string schedule starter. Values are first-pass assumptions. *)
ClearAll["Global`*"];
a4 = 440;
freqFromMidi[m_] := a4*2^((m - 69)/12);
stringTension[f_, scaleIn_, linearDensity_] := (2*scaleIn*f)^2*linearDensity;
percentBreak[scaleIn_, f_, densityLbIn3_, breakPsi_] :=
densityLbIn3*4*scaleIn^2*f^2/(breakPsi*386.4)*100;

notes = <|"C4" -> 60, "D4" -> 62, "E4" -> 64, "F4" -> 65, "G4" -> 67,
"A4" -> 69, "B4" -> 71, "C5" -> 72, "D5" -> 74, "E5" -> 76|>;
scaleLengthIn = 21;
nylonDensityLbIn3 = 0.04155;
nylonBreakPsi = 44600;

Dataset[
KeyValueMap[
<|"note" -> #1, "frequencyHz" -> N[freqFromMidi[#2]],
"percentBreakIdeal" -> N[percentBreak[scaleLengthIn, freqFromMidi[#2],
nylonDensityLbIn3, nylonBreakPsi]]|> &,
notes
]
]
```

README.md

Project artifact.

Lyre

Root-mode build packet scaffold for a small plucked lyre, aligned with the instrument-maker v4.3 Mode A packet shape.

Readiness

L2 scaffold. The packet now has a design basis, first-pass string assumptions, BOM, sourcing, cut list, validation plan, risks, photo plan, drawing brief, CAD notes, and Wolfram starter. It is not L3 because no shop prototype, measured tuning data, rendered CAD, sourceability check, or string-load proof has been completed.

Packet Contents

- `design.md` - design intent, physics model, assumptions, and known gaps.
- `bom.csv` - first-pass parts and estimated costs.
- `sourcing.csv` - source specifications and search terms.
- `cut-list.csv` - rough and final stock plan.
- `validation.csv` - tuning, load, ergonomic, and finish checks.
- `assembly-manual.md` - staged build notes for the first prototype.
- `drawing-brief.md` and `drawings/README.md` - drawing starter notes.
- `cad/README.md` - CAD starter notes.
- `wolfram-starter.wl` and `wolfram/README.md` - string model starter.
- `risks.md` - safety, structural, sourcing, and readiness risks.
- `photo-shotlist.md` - documentation image plan.

String-Scale Assumptions

Baseline: 10 strings, C4 through E5 diatonic, 21 in nominal speaking length for the mid strings, nylon or fluorocarbon monofilament, and a conservative prototype target below 70 percent of estimated breaking load. The exact gauge schedule remains a validation task.

License

photo-shotlist.md

Project artifact.

Photo Shotlist

- Orthographic front view of full-size layout before cutting.
- Close-up of grain direction through arms and yoke.
- Pin pilot-hole test block with bit sizes labeled.
- Dry-fit frame before glue.
- Bridge and nut blank before and after slotting.
- First stringing with safety setup visible.
- Tuner readings for all 10 strings after 24 hour settle.

risks.md

Project artifact.

Lyre Risks

Safety

- String breakage during first pitch-up can injure eyes or hands.
- Tuning pins may release suddenly if pilot holes are oversized.

Structural

- The yoke and arms carry the full string load and must avoid short-grain failure.
- Soundboard thickness is not validated; excessive bridge load may crown or crack the board.

Acoustic

- Gauge schedule is not measured yet, so tone balance may be uneven.
- Bridge and nut slot quality can dominate sustain and string life.

Sourcing

- Live prices, availability, and lead times are not checked.
- Fishing-line substitutions are risky unless density and breaking strength are documented.

Readiness

Remain at L2 until the CAD layout, string schedule, and first prototype validation rows are complete.